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In Vitro Activity of Twenty Commercially Available, Plant-Derived Essential Oils against Selected Dermatophyte Species

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The *in vitro* activity of twenty chemically defined essential oils (EOs) obtained from *Boswellia sacra*, *Citrus bergamia*, *C. limon*, *C. medica*, *Cinnamomum zeylanicum*, *Eucalyptus globulus*, *Foeniculum vulgare*, *Helichrysum italicum*, *Illicium verum*, *Litsea cubeba*, *Mentha spicata*, *Myrtus communis*, *Ocimum basilicum*, *Origanum majorana*, *O. vulgare*, *Pelargonium graveolens*, *Rosmarinus officinalis*, *Santalum album*, *Satureja montana*, and *Thymus serpyllum* was assayed against clinical animal isolates of *Microsporum canis*, *Trichophyton mentagrophytes*, *T. erinacei*, *T. terrestre* and *Microsporum gypseum*, main causative agents of zoonotic and/or environmental dermatophytoses in humans. Single main components present in high amounts in such EOs were also tested. Different dermatophyte species showed remarkable differences in sensitivity. In general, more effective EOs were *T. serpyllum* (MIC range 0.025%-0.25%), *O. vulgare* (MIC range 0.025%-0.5%) and *L. cubeba* (MIC range 0.025%-1.5%). *F. vulgare* showed a moderate efficacy against geophilic species such as *M. gypseum* and *T. terrestre*. Among single main components tested, neral was the most active (MIC and MFC values ≤ 0.25%). The results of the present study seem to be promising for an *in vivo* use of some assayed EOs.

Keywords: Essential oils, Dermatophytes, Antimycotic activity, Ringworm, *In vitro* sensitivity.

Superficial mycoses in both humans and animals are mainly caused by dermatophytes. These fungi are able to invade keratinized tissue and produce infections that are generally restricted to the corneous layer of skin, hair and nails. Three broad ecological groups of dermatophyte species are recognized, namely anthropophilic, zoophilic and geophilic. In general, zoophilic and geophilic species cause lesions in humans that are more inflammatory than those induced by anthropophilic species [1]. Human infection is acquired by contact with soil for geophilic species, and with infected animals or fomites for zoophilic dermatophytes.

Tinea capitis and *T. corporis*, as well as inflammatory *T. capitis* such as favus and kerion celsi (characterized by dry yellow encrustations and hair follicle suppurative infections, respectively) can be caused by zoophilic dermatophytes, such as *Microsporum canis*, *Trichophyton mentagrophytes* and *T. erinacei* [2,3]. Main animal reservoirs for the above mentioned species are cats, rodents and hedgehogs, respectively. However, a large number of pet animals can become infected and transmit dermatophytes to humans [4].

Among geophilic species, *Microsporum gypseum* is the dermatophyte more frequently involved in human mycoses. It has been associated with *Tinea corporis* [5] and with favic lesions and kerion of the scalp [2,6], while *Trichophyton terrestre* has rarely been identified as a causal agent of *Tinea capitis* [7].

In Europe *M. canis* is the most common agent for all these. Its incidence is increasing, and it is the dominant agent in Central and Southern Europe, with countries such as Austria, Spain, Italy, and Greece reporting the highest numbers and proportions of *M. canis* cases [8], even if *T. mentagrophytes* and *T. erinacei* frequently occur [3,9].

A number of anti-dermatophytic drugs are available. However, their side-effects and their decreased sensitivity lead to an extension of the treatment and can open the way to alternative care. In recent years research in aromatic and medicinal plants and particularly in their essential oils (EOs) has attracted many investigators. Several studies have shown evidence of the huge potential of these natural products as antifungal agents, justifying the current use of these compounds in a number of pharmaceutical, food and cosmetic products [10]. Therefore, EOs are considered promising natural products for the development of broad-spectrum, safe and cheap antifungal agents.

Furthermore, EOs have been used in veterinary medicine both as monotherapy and associated in mixtures with good outcomes, sometimes better than results from conventional treatments [11-13]. The aim of the present paper was to evaluate the *in vitro* activity of 20 chemically defined EOs against isolates of *M. canis*, *T. mentagrophytes*, *T. erinacei*, *M. gypseum* and *T. terrestre* obtained from dermatologically diseased animals.

The choice of the EOs was made on the basis of our previous studies and on their availability on the market. The chemical composition of the tested essential oils is reported in Table I. Their aromatic profile is quite different since the plant material belongs to different botanical families. In the majority of the samples there is a good amount of monoterpene hydrocarbons and oxygenated monoterpenes, with the exception of *C. limon* and *C. medica*, where monoterpene hydrocarbons are the predominant class of compounds (92.5% and 96.2%), with limonene as the main compound (59.2% and 92.2%, respectively). *I. verum* is characterized by high amount of phenyl propanoids (94.2%, due exclusively to (*E*)-anethol), also present in *C. zeylanicum*, even though in a lower amount (80.1%). *P. graveolens* and *E. globulus* show a high percentage of oxygenated monoterpenes (86.2 % and 90.5%,

Table 1: Chemical composition of the tested EOs (main compounds more than 1%)*.

Compounds	LRI§	L.c.	M.s.	S.a.	P.g.	I.v.	C.z.	F.v.	C.b.	O.m.	O.v.	S.m.	E.g.	H.i.	M.c.	R.o.	T.s.	O.b.	C.l.	C.m.	B.s.
α-Thujene	932													7.2							54.2
Tricyclene	938														49.0						
α-Pinene	940	1.1						2.2			1.0		2.0			37.9			2.3		6.2
α-Fenchene	951													1.2							
Camphene	955															5.4					
Thuja-2,4(10)-diene	959																				7.3
Sabinene	978	1.0								3.2									2.5		
β-Pinene	981	0.9	1.0						2.8					1.00		5.0			13.7		1.1
6-Methyl-5-hepten-2-one	990	1.4																			
Myrcene	993	1.1								1.6	2.2	1.0				1.6			1.8	2.2	
α-Phellandrene	1006							1.3													3.7
α-Terpinene	1019									4.7	2.1	1.2									5.1
o-Cymene	1026															1.1					3.3
p-Cymene	1028						1.0	1.7	3.8	4.2	9.3	9.0	6.7	1.1	2.7		15.3				
Limonene	1032	11	1.7			2.9		1.8	27.1	2.1				7.0	5.9	3.3			59.2	92.2	
1,8-Cineole	1036		5.2										89.8	2.3	29.0	22.0		5.9			
γ-Terpinene	1062								3.5	7.9	5.3	6.1					2.9		10.8		
cis-Sabinene hydrate	1072									3.2											
Terpinolene	1090									1.5											
Fenchone	1090							18.4													
trans-Sabinene hydrate	1098									12.8	1.8						3.8				
Linalool	1102	1.7			3.9				16.8			3.1			1.5			46.0			
cis-Thujone	1108																				1.1
Sabinol	1143																				1.5
Camphor	1148															7.6					
Menthone	1154		16.4		1.1																
Citronellal	1155	1.7																			
Isomenthone	1164				3.5																
neo-Menthol	1166		11.2																		
Borneol	1169											2.1					1.6				
Menthol	1178		39.0																		
4-Terpineol	1180									17.6							2.4				
α-Terpineol	1192									2.7											
unknown																	1.7				
Methyl chavicol (= Estragol)	1198							2.9										1.1			
Citronellol	1231				44.5																
Thymol methyl ether	1232																1.7				
Pulegone	1237		3.3																		
Isobornyl formate	1239																				2.5
Neral	1242	32.0																	1.0		
Carvone	1248											1.9									
p-Anisaldehyde	1258							1.9													
Geraniol	1259	1.5			13.7					2.7											
Linalool acetate	1260								34.5	3.2		1.2									
Geranial	1276	37.0																	1.5		
Citronellyl formate	1280				7.3																
(*E*)-Anethole	1283					94.2		65.0													
Isobornyl acetate	1285															3.3		1.6			
Thymol	1290											2.6					52.6				
Safrole	1291						1.4														
Menthyl acetate	1294		6.9																		
Geranyl formate	1298				1.9																
Carvacrol	1299									20.8	65.9	47.1									
Eugenol	1361						64.8											11.5			
Neryl acetate	1368													31.8					1.0		
β-Elemene	1392																	2.2			
β-Caryophyllene	1418	2.2	2.1				2.5			1.7	3.7	3.6		3.1		4.1	6.8				
trans-α-Bergamotene	1437																	3.6			
α-Guaiene	1440																	1.1			
Aromadendrene	1445				2.9																
Cinnamyl acetate	1446						1.1														
unknown														2.6							
Isoeugenol	1449						14.0														
α-Humulene	1456											1.6									

Neryl propanoate	1454													5.1							
β-Santalene	1461			1.4																	
Germacrene D	1481													3.4				3.5			
ar-Curcumene	1484													5.6							
trans-Muurola-4(14),5-diene (=Bicyclosesquiphellandrene)	1492													2.1							
Valencene	1493													1.7							
Bicyclogermacrene	1495																	1.0			
α-Muurolene	1499									1.4											
α-Bulnesene	1505																	2.0			
β-Bisabolene	1509											1.5							1.1		
trans-γ-Cadinene	1513													1.1				2.8			
δ-Cadinene	1523											1.4					1.0				
Citronellyl butyrate	1532				1.2																
Selina-3,7(11)-diene	1542			1.0																	
Elemol	1553			4.7																	
Geranyl butyrate	1564				1.1																
(*E*)-2-Phenyl ethyl tiglate	1585				1.2																
5-*epi*-7-*epi*-α-Eudesmol	1606													5.5							
epi-10-γ-Eudesmol	1627			4.9																	
γ-Eudesmol	1634			5.2																	
τ-Cadinol	1640																	5.8			
β-Eudesmol	1649			5.5																	
(*Z*)-Citronellyl tiglate	1658				1.2																
Valerianol	1658			14.4																	
7-*epi*-α-Eudesmol	1664			5.9																	
(*Z*)-α-Santalol	1672			27.1																	
(*Z*)-*trans*-α-Bergamotol	1691			2.0																	
Geranyl tiglate	1696				1.2																
(Z)-β-*cis*-Santalol	1705			2.1																	
(*Z*)-β-*trans*-Santalol	1710			10.8																	
Benzyl benzoate	1764						2.7														
(*Z*)-Lanceol	1768			1.2																	
Monoterpene hydrocarbons		4.4	7.1	-	-	3.0	3.3	8.2	38.8	25.4	34.6	19.8	9.1	19.2	61.2	56.5	21.7	2.0	92.5	96.2	84.3
Oxygenated monoterpenes		50.4	76.7	-	86.2	-	1.3	19.4	57.7	63.0	58.1	62.1	90.5	35.5	31.1	36.7	64.1	56.1	4.9	0.5	6.8
Sesquiterpene hydrocarbons		39.6	10.6	5.6	7.8	-	3.6	-	1.6	4.9	5.8	11.9	0.3	29.4	1.2	4.4	9.2	20.0	2.6	1.9	3.7
Oxygenated sesquiterpenes		0.3	1.5	88.8	4.1	-	0.4	-	0.5	0.9	0.2	1.1	0.1	9.4	-	0.3	-	7.9	-	-	-
Phenyl propanoids		-	-	-	-	94.2	80.1	67.9	-	-	-	0.2	-	-	-	-	-	12.7	-	-	0.2
Others		1.4	0.2	-	1.2	-	4.1	1.9	0.1	0.2	1.0	0	-	-	-	-	0.8	0.2	-	0.6	-
Unknowns		-	-	-	-	-	-	-	-	-	-	-	-	2.6	0.3	0.1	1.7	-	-	-	-
Total % of chemical classes		96.8	95.9	94.4	99.2	97.2	92.8	97.4	98.6	98.2	98.4	95.0	99.9	94.2	93.8	97.9	97.3	99.2	100.0	99.2	95.0

* LRI§: on DB-5 column. *L.c.* (*Litsea cubeba*), *M.s.* (*Mentha spicata*), *S.a.* (*Santalum album*), *P.g.* (*Pelargonium graveolens*), *I.v.* (*Illicium verum*), *C.z.* (*Cinnamomum zeylanicum*), *F.v.* (*Foeniculum vulgare*), *C.b.* (*Citrus bergamia*), *O.m.* (*Origanum majorana*), *O.v.* (*Origanum verum*), *S.m.* (*Satureja montana*), *E.g.* (*Eucalyptus globulus*), *H.i.* (*Helichrysum italicum*), *M.c.* (*Myrtus communis*), *R.o.* (*Rosmarinus officinalis*), *T.s.* (*Thymus serpyllum*), *O.b.* (*Ocimum basilicum*), *C.l.* (*Citrus limon*), *C.m.* (*Citrus medica*), *B.s.* (*Boswellia sacra*)

respectively). Only the EO of *S. album* among the others is rich in oxygenated sesquiterpenes (88.8%). However, the chemical composition of tested EOs was substantially consistent with the literature data, except for *L. cubeba*. In fact the EO used in the present study, although containing lower amounts of neral, geranial [14] and citral [15] in comparison with the published data, showed good antidermatophyte activity. To the best of our knowledge, there is only one paper dealing with the anti *M. canis* activity of this EO [20], whilst there is no report about its efficacy against other dermatophyte species.

The selected EOs showed a variable degree of antimycotic activity at the tested dilutions, with MICs ranging from 0.025% to > 10%. MIC and MFC values varied among the different dermatophytic species tested. In general terms, the most effective EOs were *T. serpyllum*, *O. vulgare* and *L. cubeba* which had an overall MIC range of 0.025% to 0.25%, 0.025% to 0.5% and 0.025% to 1.5%, respectively. *F. vulgare* showed a moderate efficacy against geophilic species such as *M. gypseum* and *T. terrestre.* Among all the tested fungal species, *M. canis* had the lowest MIC and MFC values.

Among conventional drugs tested, terbinafine was found to be the most effective against *M. canis*, *M. gypseum*, *T. terrestre* and *T. erinacei*, while the best results against *T. mentagrophytes* were achieved using voriconazole. More detailed data are reported in Table 2.

Neral was the most effective compound among the single main components tested, showing MIC and MFC values ≤ 0.25%. This compound was significantly present only in *L. cubeba* EO. Thymol, carvacrol, eugenol, geranial, geraniol and fenchone showed satisfactory activity against the majority of the tested fungal species. Limonene, *p*-cymene, α-pinene and γ-terpinene did not yield any antifungal effect at 10%. Results are reported in detail in Table 3.

The overall sensitivity to EOs of the examined fungal species showed some marked differences except for zoophilic *Trichophyton* species (*T. mentagrophytes* and *T. erinacei*), which showed quite an homogeneous overall sensitivity pattern. To the best of our knowledge there are no previous studies including such animal/human pathogens.

Table 2: Minimum inhibitory concentrations (MIC) and minimum fungicidal concentrations (MFC) of twenty essential oils and MIC of conventional antimycotic drugs when tested against selected dermatophytes.

	Microsporum canis		*Microsporum gypseum*		*Trichophyton mentagrophytes*		*Trichophyton terrestre*		*Trichophyton erinacei*	
EOs	**MIC**	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**
Litsea cubeba	0.025	0.25	0.25	0.5	0.25	0.5	1.5	5	0.25	2.5
Foeniculum vulgare	0.25	1.5	0.5	1.5	1.5	2	1.5	3	3	3
Illicium verum	3	7.5	1	2	2	2.5	1.5	5	3.5	5
Pelargonium graveolens	0.25	0.5	1.5	2.5	0.75	2	0.75	3	0.75	2
Eucalyptus globulus	0.1	0.5	0.5	0.5	0.5	1	1.5	3	0.5	0.5
Satureja montana	0.5	3	2	5	2	7.5	3	7.5	2	7
Origanum majorana	0.5	4	2	7.5	1	6	2	7.5	1.5	6
Menta spicata	2	5	3	7.5	3	5	3	5	3	5
Santalum album	7.5	>10	7,5	>10	7.5	>10	10	>10	7.5	>10
Cinnamomum zeylanicum	4.5	5	7,5	10	7.5	>10	10	>10	7.5	>10
Citrus bergamia	4	5	5	7.5	5	7.5	5	7.5	7.5	>10
Boswellia sacra	5	>10	7.5	>10	7.5	>10	10	>10	8	>10
Origanum vulgare	0.025	0.05	0.025	0.05	0.5	0.5	0.25	1.5	0.5	1
Thymus serpyllum	0.025	0.05	0.25	0.5	0.1	0.1	0.1	1	0.2	0.5
Helicrysum italicum	5	>10	10	>10	10	>10	10	>10	10	>10
Myrtus communis	2	3	3	5	1,5	5	3	6	2	5
Ocimum basilicum	1	5	3	5	2.5	5	3	6	2.5	5
Citrus medica	4	5	5	7.5	7.5	10	10	10	8	10
Citrus limon	2.5	7.5	2.5	10	5	7.5	7.5	10	5	7.5
Rosmarinus officinalis	2.5	7.5	2.5	7.5	5	7.5	5	7.5	1.5	1.5
Synthetic drugs	mg/L		mg/L		mg/L		mg/L		mg/L	
Griseofulvin	1		40		160		40		2	
Terbinafine	0.0156		0.16		16		0.16		0.01	
Itraconazole	0.125		32		32		0.8		0.25	
Posaconazole	0.0625		8		16		1.6		8	
Voriconazole	0.0625		3.2		3.2		2		0.25	

Table 3: Minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC) of main compounds present in the more effective essential oils tested against examined dermatophytes.

	Microsporum canis		*Microsporum gypseum*		*Trichophyton mentagrophytes*		*Trichophyton terrestre*		*Trichophyton erinacei*	
	MIC	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**	**MIC**	**MFC**
τ-Anethol	1%	5%	1%	> 10%	2.5%	> 10%	2.5%	> 10%	2.5%	> 10%
Carvacrol	0.1%	1%	0.25%	0.5%	0.06%	4%	0.1%	2.5%	0.1%	2.5%
ρ-Cymene	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%
1,8-Cineole	5%	> 10%	5%	> 10%	1%	> 10%	10%	> 10%	1%	> 10%
Neral	0.1%	0.25%	0.1%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%
Eugenol	0.1%	0.25%	0.25%	0.25%	0.25%	0.5%	0.25%	1%	0.25%	1%
Geraniol	0.25%	0.25%	0.25%	0.5%	1%	1%	0.5%	1%	0.5%	1%
Geranial	0.1%	0.25%	0.1%	0.5%	0.25%	1%	1.5%	5%	0.25%	2%
Citronellol	0.25%	0.25%	0.25%	0.5%	0.25%	0.5%	0.5%	2%	0.5%	1.5%
Limonene	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	>10%	>10%
Linalool	1%	2.5%	1%	5%	2.5%	10%	2.5%	10%	2.5%	10%
Menthol	0.5%	5%	2.5%	5%	1%	> 10%	1%	10%	1%	> 10%
Menthone	1%	> 10%	1%	> 10%	2.5%	> 10%	1%	> 10%	2.5%	> 10%
α-Pinene	10%	> 10%	10%	> 10%	> 10%	> 10%	10%	> 10%	10%	> 10%
γ-Terpinene	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	> 10%	10%	> 10%
Thymol	0.05%	0.5%	0.25%	0.5%	0.125%	1%	0.25%	2.5%	0.25%	2.5%
Fenchone	0.25%	0.5%	0.5%	0.5%	1%	4%	1%	4%	1.50%	4%

T. serpyllum (rich in thymol, 52.6%) presented the lowest MIC value against all fungi, except for *M. gypseum*. For this last species *O. vulgare* gave the best results both in terms of MIC and of MFC. Furthermore *T. serpyllum* showed the best results when tested against *Trichophyton* species. This feature would be due to the large amounts of thymol that has antifungal efficacy in concentrations ranging from 0.05% to 2.5%. Both *Microsporum* species and zoophilic *Trichophyton* were sensitive to *L. cubeba* EO, probably due to the large amounts of neral and geranial. *O. vulgare* had good antimycotic activity against the *Microsporum* species tested, confirming the good efficacy of carvacrole.

T. terrestre showed a peculiar sensitivity pattern; it was sensitive to *T. serpyllum* even though the MFC value obtained was quite high in respect of other examined fungi. The most striking finding was the low sensitivity to *L. cubeba* EO, probably due to the poor activity of geranial versus this fungal species.

The MIC values obtained with conventional antimycotic drugs were in substantial agreement with the literature data [17-20]. Several EOs were tested against both geophilic and zoophilic fungi; these latter have been chosen on the basis of their broad host range and their zoonotic potential. Our data confirm anti dermatophyte efficacy of *T. serpyllum, O. vulgare* [21, 22] and *L. cubeba* EOs [16].

The results of the present study seem to be promising for a possible *in vivo* use of these EOs, alone and/or as a mixture. The topical administration of EO mixtures appeared to be effective in veterinary medicine [12-13] allowing the use of lower concentrations of each component and showing better antimycotic activity in comparison with the ingredients used alone.

The use of these natural products could be advisable for topical administration in combination with systemic treatment. In particular, *T. mentagrophytes* appeared insensitive to all the synthetic antimycotic drugs, except voriconazole, while MIC and MFC values for *T. serpyllum*, *L. cubeba* and *O. vulgare* demonstrated good antimycotic efficacy. This finding was partially referable to *M. gypseum* also, being fully sensitive to terbinafine and voriconazole only, and inhibited by low concentrations of *O. vulgare*, *L. cubeba*, *T. serpyllum* and *F. vulgare* EOs.

Differences in drugs/remedia sensitivity when tested against distinct dermatophyte species were remarkable. For this reason, it is important to reiterate the capital role of the correct identification of the etiological agents of ringworm, in order to set up the proper treatment. Some of the tested EOs were highly effective against dermatophytes. Further clinical studies are needed to evaluate the possibility of applying these natural products to the treatment of dermatophytosis to provide an effective and safe additional therapeutic option.

Experimental

Essential Oils: *Boswellia sacra*, *Citrus bergamia*, *C. limon*, *C. medica*, *Cinnamomum zeylanicum*, *Eucalyptus globulus*, *Foeniculum vulgare*, *Helicrysum italicum*, *Illicium verum*, *Litsea cubeba, Mentha spicata*, *Myrtus communis*, *Ocimum basilicum*, *Origanum majorana*, *O. vulgare*, *Pelargonium graveolens*, *Rosmarinus officinalis*, *Santalum album*, *Satureja montana* and *Thymus serpyllum* EOs were employed for all *in vitro* studies. EOs (20% solutions) were prepared in sweet almond oil (*Prunus dulcis* Mill. Flora Srl., Lorenzana, Pisa, Italy). The above mentioned EOs were supplied by Flora srl (Lorenzana, Pisa, Italy).

GC-MS analysis: Volatile constituents of each EO were analyzed by GC-MS, as previously reported [23]. Briefly, a CP-3800 gas chromatograph equipped with HP-5 capillary column (30 m X 0.25 mm; coating thickness, 0.25 mm) and Varian Saturn 2000 ion trap mass detector were employed. Analytical conditions were as follows: injector and transfer line temperature, 220 and 240°C respectively; oven temperature, programmed from 60 to 240°C at 3°C/min; carrier gas, helium at 1 mL/min; injection, 0.2 mL (10% *n*-hexane solution); split ratio, 1:30. Identification of the constituents was based on comparison of the retention times with those of authentic samples, comparing their linear retention indices relative to a series of *n*-hydrocarbons, and on computer matching against commercial and home-made library mass spectra built up from pure substances and components of known oils and MS literature data [24,25].

In vitro assay: The *in vitro* antimycotic activity of EOs was evaluated on clinical isolates of *M. canis*, *T. mentagrophytes*, *T. erinacei*, *M. gypseum* and *T. terrestre*, respectively. The dermatophytes were cultured by affected cats (*M. canis* and *T. mentagrophytes*) and dogs (*T. erinacei*, *M. gypseum* and *T. terrestre*) and maintained on Sabouraud Dextrose Agar (SDA). The effectiveness of EOs was assessed by means of a microdilution test carried out as previously described using a semisolid malt extract medium (MEA) with 1% agar [26]. Portions of approximately 1 mm^3 non-sporulating mycelia from SDA were used as fungal inocula in 24-well plates (Pbi International, Milano, Italy). Stock solutions at 20% of all the chemically defined EOs were diluted in culture medium to obtain concentrations ranging from 10% to 0.01%.

Control cultures were achieved using medium alone and medium supplemented with 0.5% sweet almond oil. Plates were incubated at 25°C for about 10 days or until the full development of mycotic growth in control wells was observed. Portions of the inocula where growth was not visually noticed were removed, twice washed in sterile saline and then seeded onto solid MEA plates to evaluate the viability of the fungi. All tests were performed in quadruplicate. Further controls were achieved using griseofulvin (Sigma Aldrich, Italy), itraconazole (Janssen Cilag, Italy) terbinafine (Sandoz, Italy), voriconazole (Sigma Aldrich, Italy) and posaconazole (Sigma Aldrich, Italy) at concentrations from 160 mg/kg to 0.02 mg/kg following the procedure described by Mancianti *et al*. [27].

Single main components present in high amounts in EOs were separately tested *in vitro*, as well as at concentrations ranging from 10% to 0.02%. So anethol, carvacrol, *p*-cymene, 1,8 cineole, thymol, eugenol, citronellol, geranial, neral, geraniol, fenchone, linalool, menthol, menthone, α-pinene, γ-terpinene and limonene were assayed against all dermatophytes, as described above. All tests were carried out in quadruplicate. All these components were provided by Sigma-Aldrich (Italy). The viability of the inocula was evaluated as described above.

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